Deployment environments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Deployment environments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Queues and asynchronous work

Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Data retention and deletion

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

External side effects and recovery

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production.

For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed **environment** names. A staging run cannot use production write credentials.

Practise with fading help

> Complete the supported Deployment environments trace and record the first step you can explain without help.

Use W2: Trace the worked case for Deployment **environment**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Deployment environments explanation against the supplied facts before discussing it.

Use the worked case for Deployment **environment**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Development uses fake records but a production token. What boundary is violated?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Deployment environments facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Deployment environments facts and keep your reasoning.

Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Deployment environments facts and keep your reasoning.

A report omits this qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Deployment environments facts and keep your reasoning.

Development uses fake records but a production token. What boundary is violated?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Deployment environments facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Deployment environments to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled deployment **environment**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of deployment **environment**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Deployment environments

> Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records. Explain the labelled step before checking the changed condition.

Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

Staging card names trial room Z; production card names room A. Credentials and stores are separate.

AI ACADEMY / AI-6/23

Deployment environments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Use staging data

The rehearsal reads fictional room Z.



2. Check actual connection

Staging credentials must not reach production storage.



3. Review evidence

A passing rehearsal supports release review, not automatic promotion.

CHANGE ONE THING

Point staging at the production credential.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A passing rehearsal supports release review, not automatic promotion.

1. Use staging data

2. Check actual connection

3. Review **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review **evidence** A passing rehearsal supports release review, not automatic promotion. An **environment** includes code, configuration, permissions and data.

A staging label alone cannot create separation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Point staging at the production credential. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Stop the rehearsal because its separation boundary is breached.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

Repair a missing building block

Review Queues and asynchronous work (ai-6/22). Goal: Explain the **mechanism** and terminology of queues and asynchronous work. Short practice: Explain what the queued J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

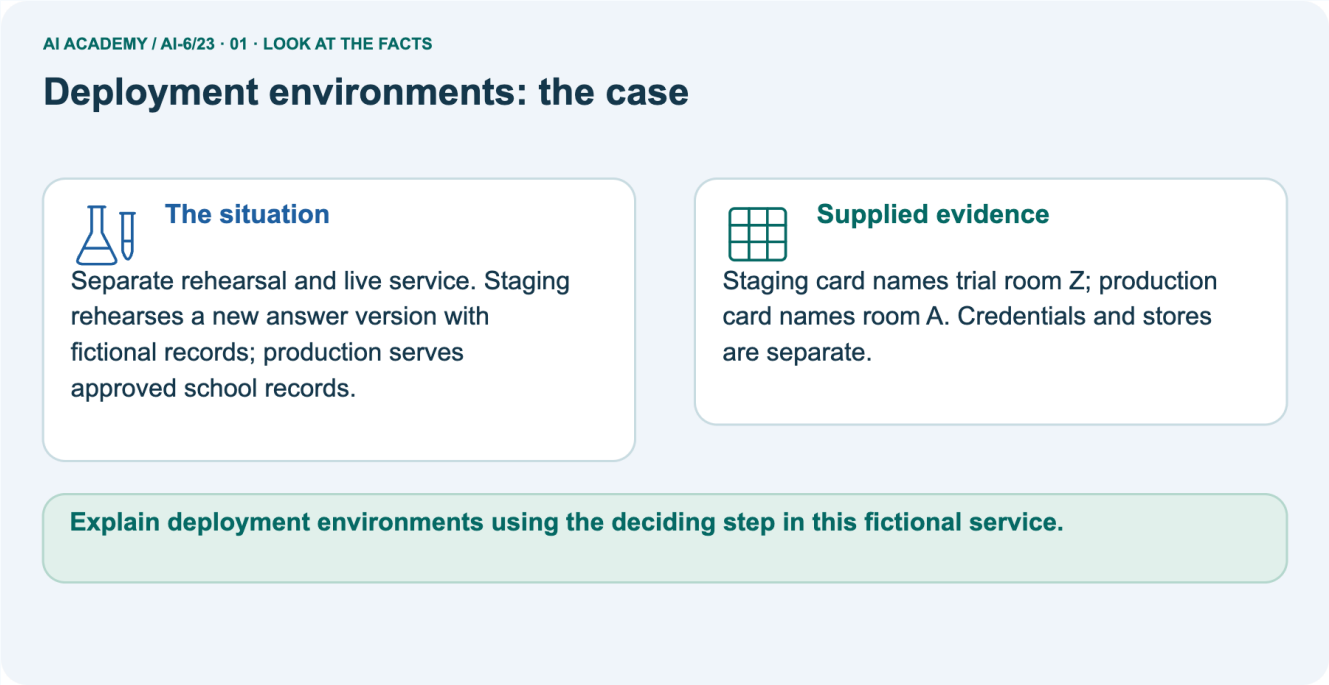
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Deployment environments picture from its labels, then record a relevant limit.

01. Deployment environments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

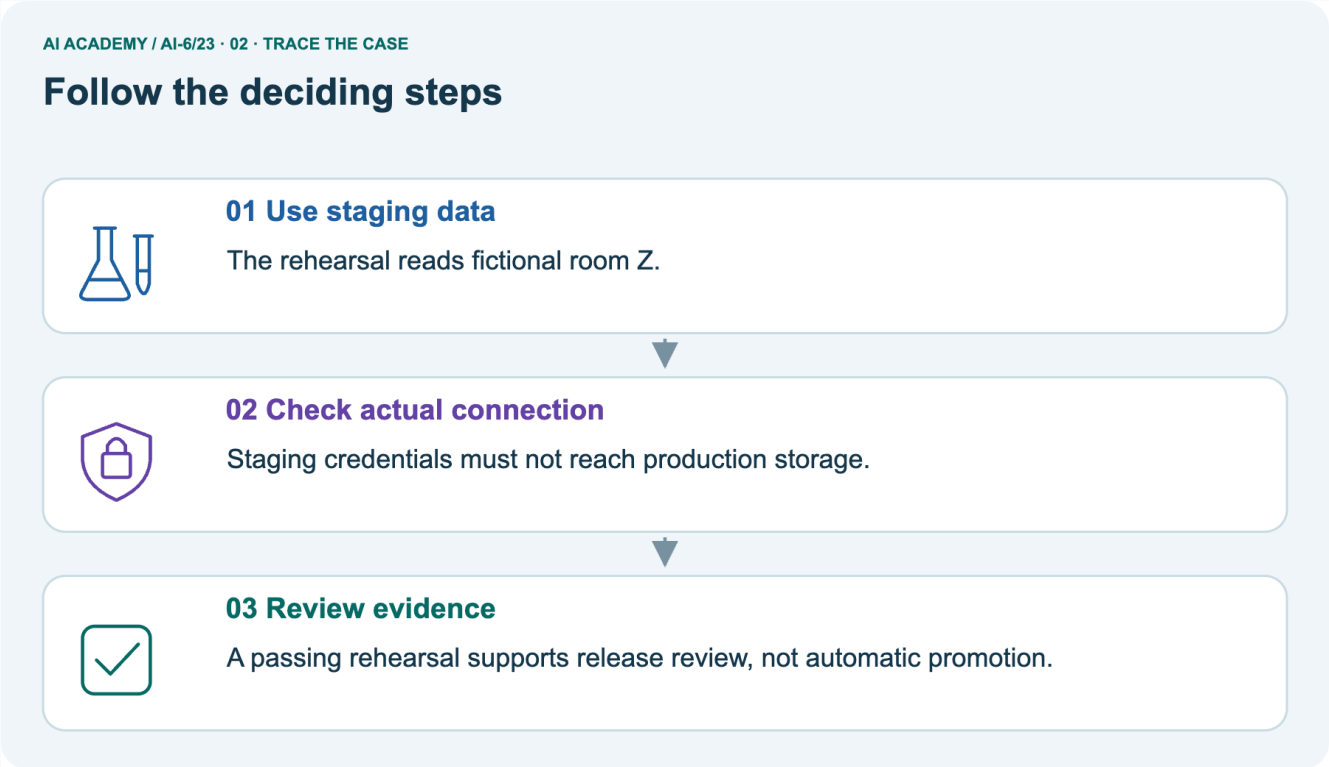
Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records. Supplied facts: Staging card names trial room Z; production card names room A.

Credentials and stores are separate.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



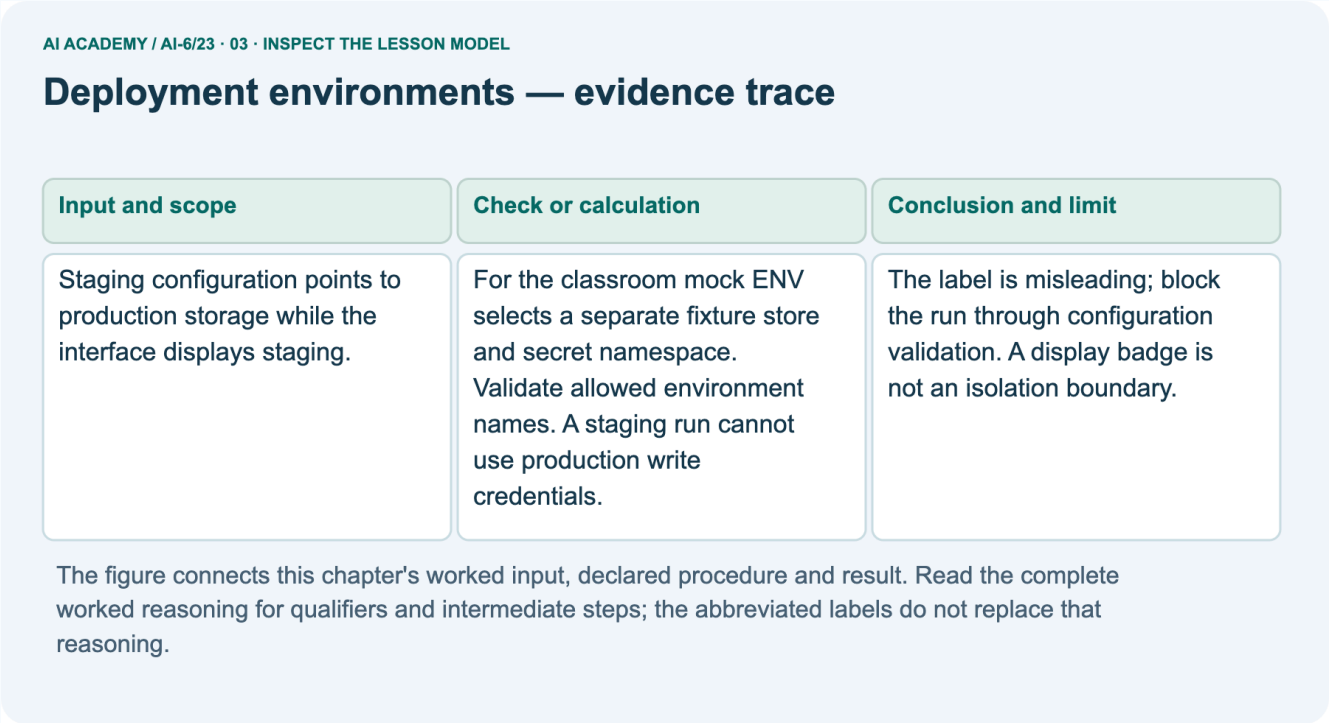
Follow the labelled picture with the matching explanation. Open the vector diagram

Use staging data: The rehearsal reads fictional room Z. → Check actual connection: Staging credentials must not reach production storage. → Review **evidence**: A passing rehearsal supports release review, not automatic promotion.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Deployment environments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

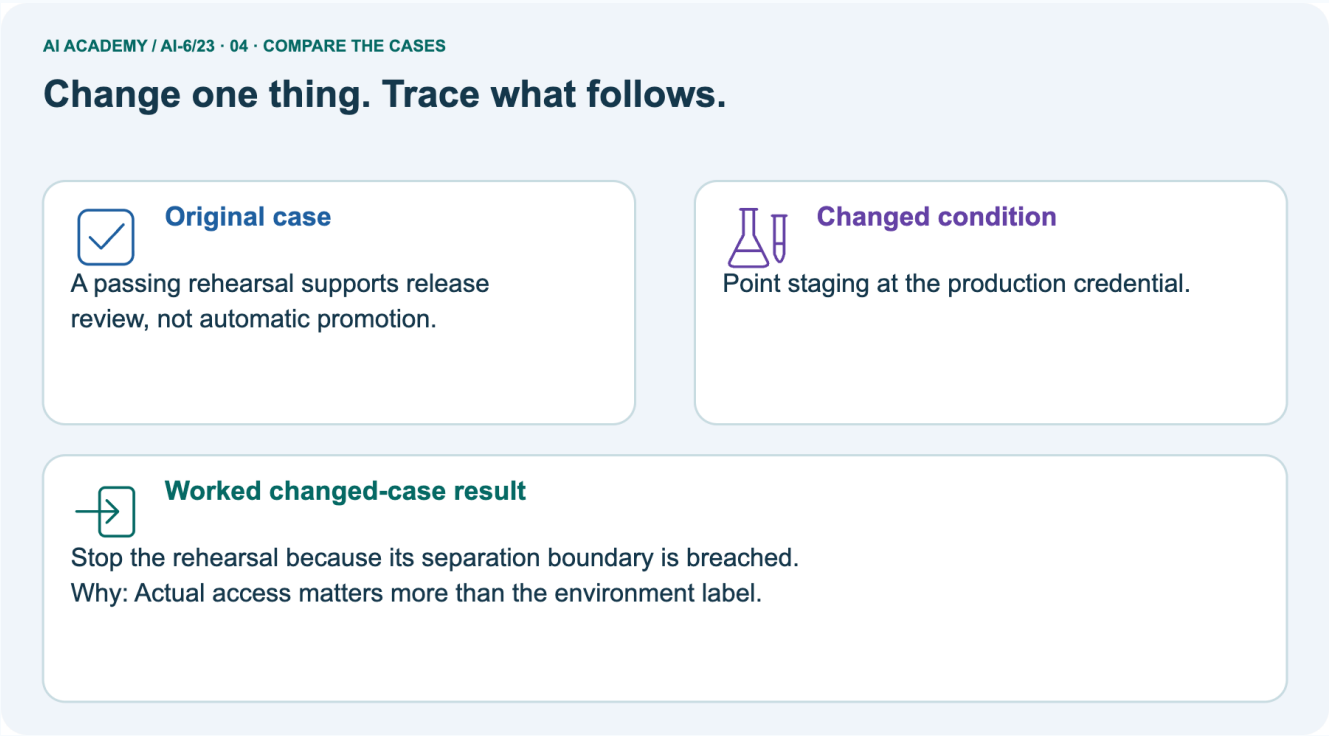
Deployment **environment**s — **evidence** trace. Staging configuration points to production storage while the interface displays staging. | For the classroom mock ENV selects a separate fixture store and secret namespace.

Validate allowed **environment** names. A staging run cannot use production write credentials. | The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

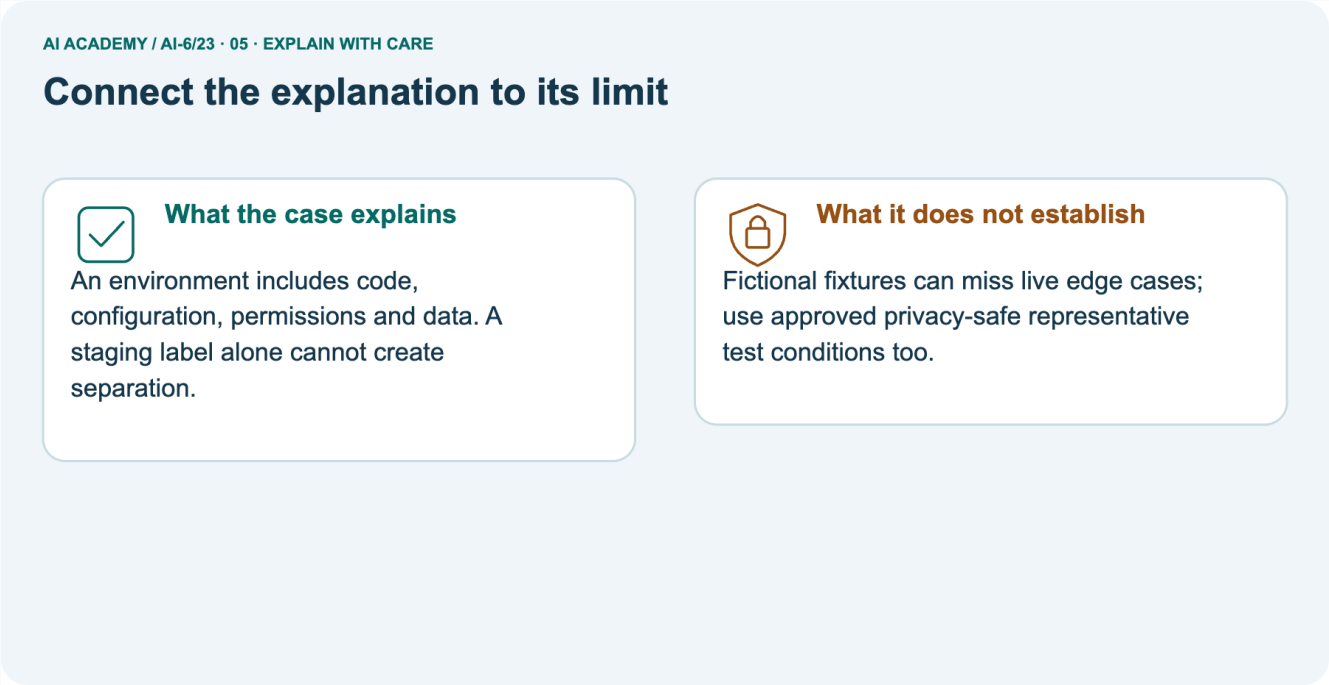
Original case: A passing rehearsal supports release review, not automatic promotion. Changed condition: Point staging at the production credential.

Expected result: Stop the rehearsal because its separation boundary is breached. Reason: Actual access matters more than the **environment** label.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

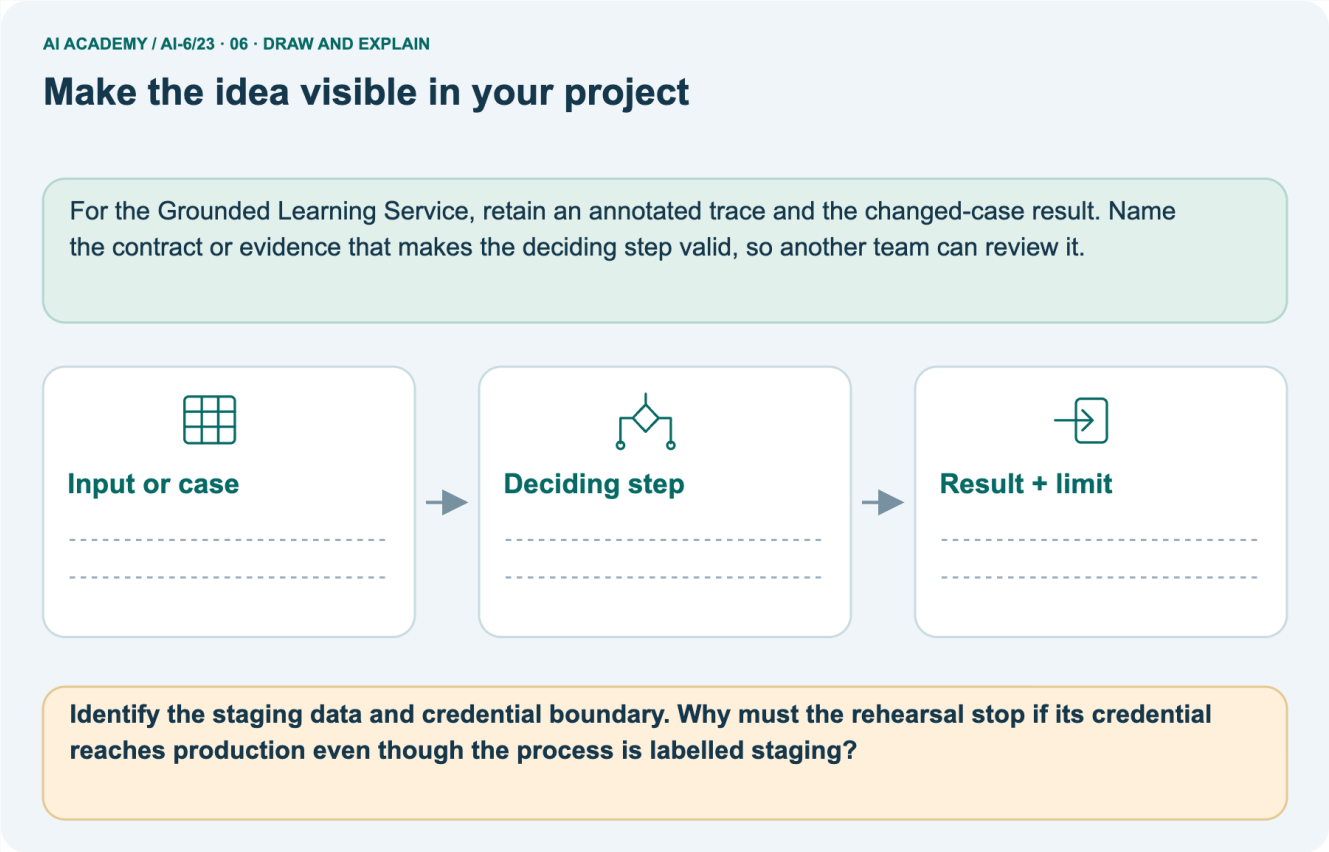
An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

Limit: Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.